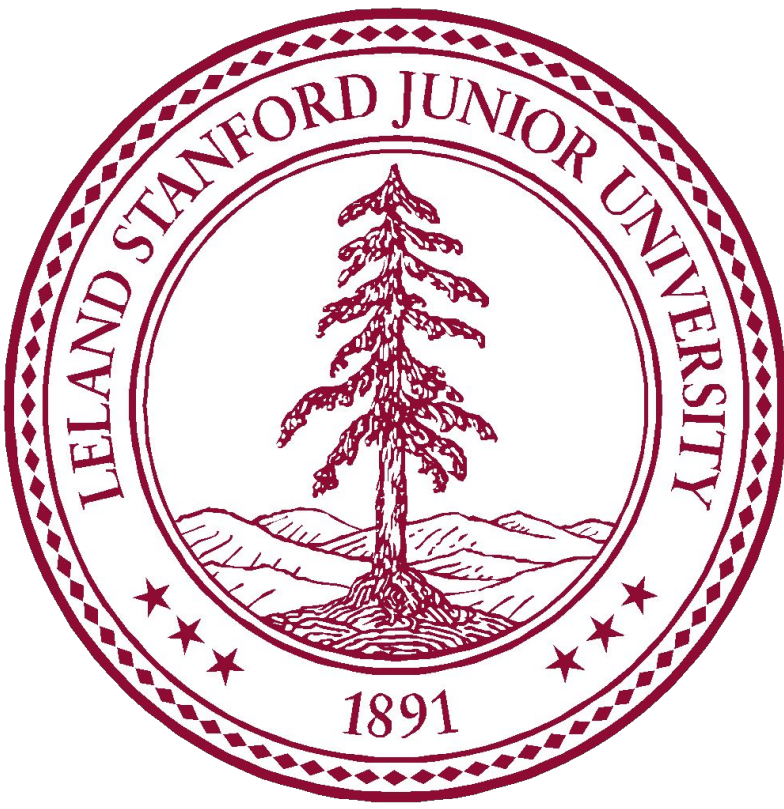




Combining Delivery of Insulin and GLP-1 Agonists to Optimize Medication Administration in Type 2 Diabetes Patients

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Introduction

- 529 million people live with diabetes worldwide. Specifically, **36 million** people in the US have Type 2 diabetes (T2D), meaning approximately **1 in 10 people** in the US have T2D¹.
- T2D occurs due to defective insulin secretion by pancreatic β -cells, where insulin-sensitive tissues are unable to respond to insulin, causing **insulin resistance** and **increased blood glucose levels**.
- Systems of administration for patients prescribed with GLP-1 Receptor Agonists (GLP-1RAs) failed **50.9%** of the time³.
- Missing medication for T2D patients can lead to **life-threatening** health complications and **increased** medical costs.
- Our product **streamlines** the medication administration process by **combining** the delivery of **insulin** and **GLP-1RA** into one **automated device**.

Need Statement

A way to optimize medication administration systems in T2D patients whose treatment includes both insulin and GLP-1RAs to increase time in optimal blood glucose (HbA1c) levels.

Need Specifications

Feature:	Need to have:	Nice to have:
Dosage Accuracy	HbA1C levels < 7%, 70% of the time ⁴	HbA1C levels < 7%, 95% of the time
User-Friendliness	< 5 mins per usage	< 3 mins per usage
Weight	< 1 oz	< 0.5 oz
Size	< 20 cubic inches	< 8 cubic inches
Affordability	< \$150 ⁵ per unit	< \$100 per unit

References

¹ CDC. "National Diabetes Statistics Report." Diabetes, June 6, 2024. <https://www.cdc.gov/diabetes/php/data-research/index.html>.

² Mayo Clinic. "Type 2 Diabetes - Symptoms and Causes." Accessed July 7, 2024. <https://www.mayoclinic.org/diseases-conditions/type-2-diabetes/symptoms-causes/syc-20351193>.

³ "Real-World Adherence and Discontinuation of Glucagon-Like Peptide-1 Receptor Agonists Therapy in Type 2 Diabetes Mellitus Patients in the United States." NCBI, 27 November 2020. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7708309/>. Accessed 17 July 2024.

⁴ McMillen, Matt. "Hemoglobin: A1C Test, Range, and Normal Levels." WebMD, 18 June 2024. <https://www.webmd.com/diabetes/glycated-hemoglobin-test-hba1c>. Accessed 25 July 2024.

⁵ Wan, Wen. "Cost-effectiveness of Initiating an Insulin Pump in T1D Adults Using Continuous Glucose Monitoring Compared with Multiple Daily Insulin Injections: the Diamond Randomized Trial." NCBI, <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6226055/>. Accessed 26 July 2024.

Value Proposition

Current Standard of Care:

Switch insulin pump or Omnipod every 3 days

Get GLP-1RA injections once a week

Our Product:

Prefilled cartridges with both insulin and GLP-1, switch out our product every 3 days

Solution Design

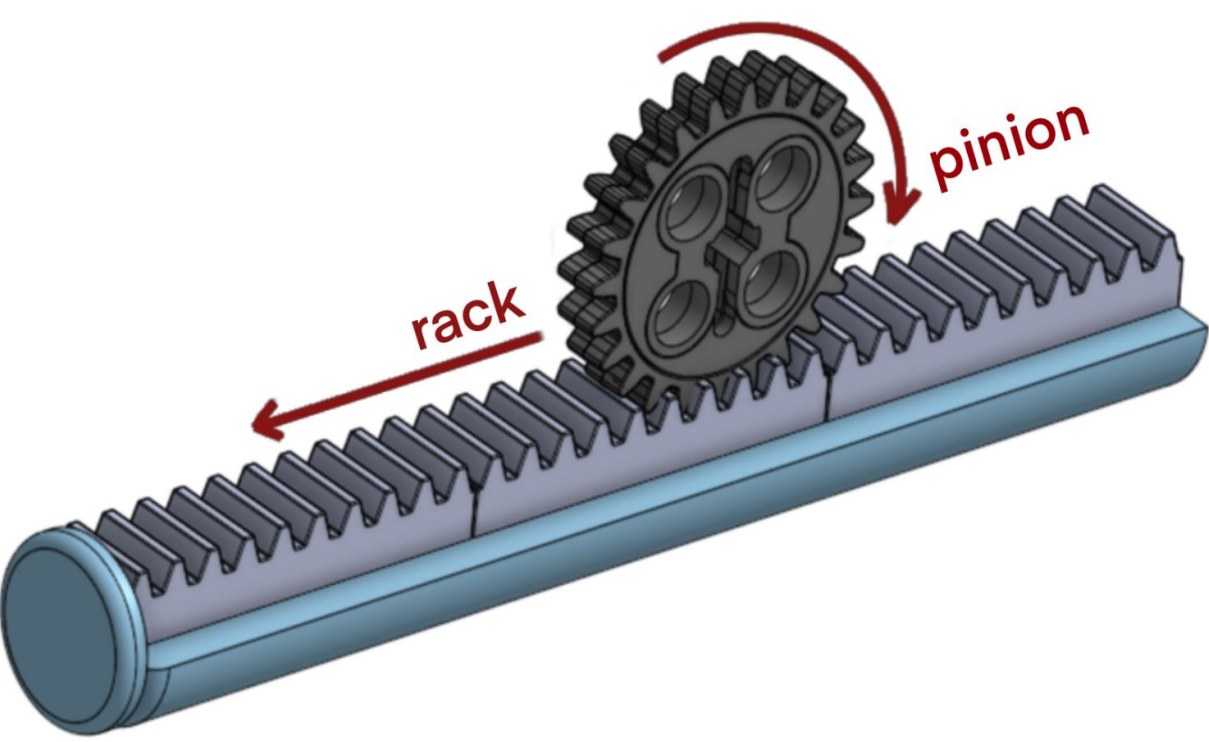


Figure 1: Rack and pinion system converts rotational motion to translational motion

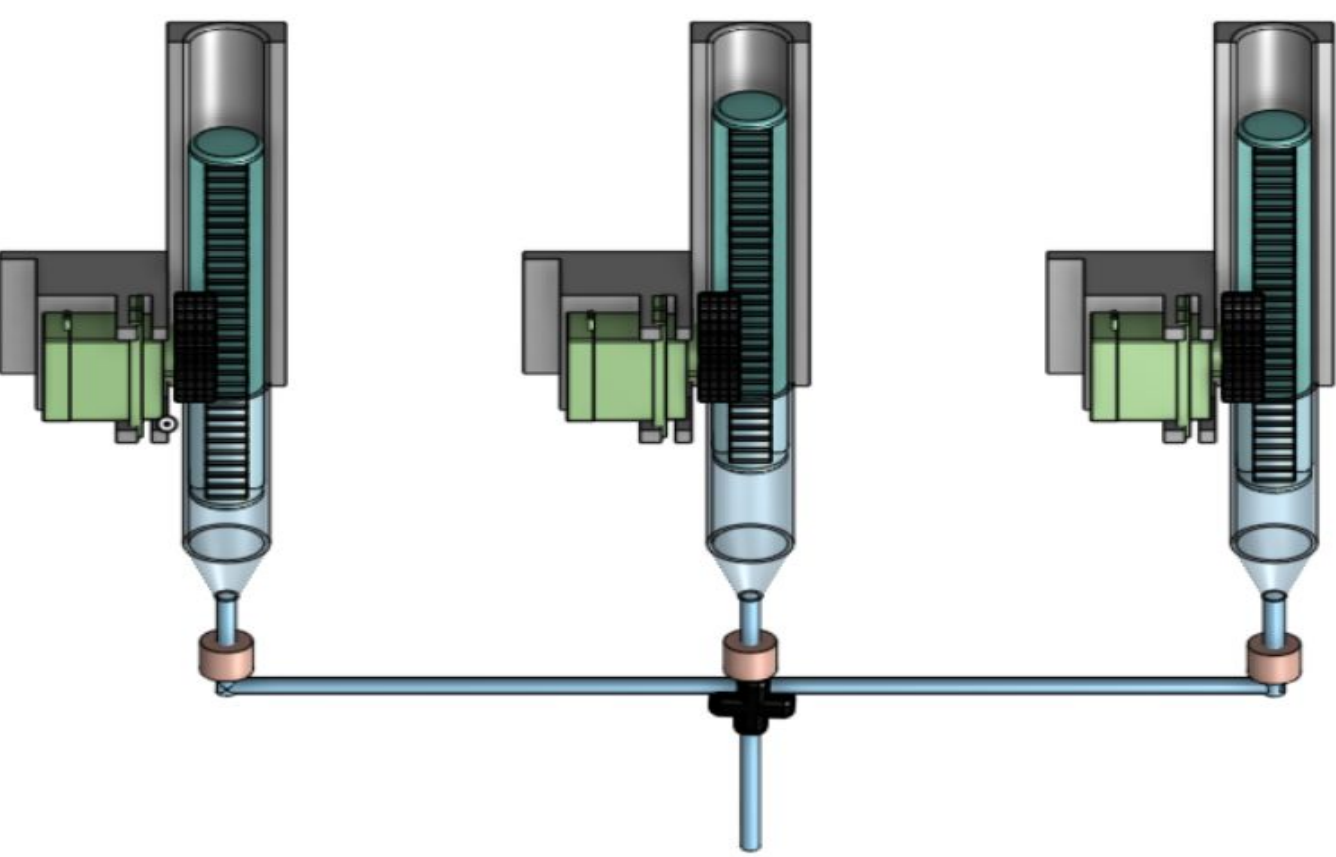


Figure 2: CAD prototype to deliver insulin and GLP-1

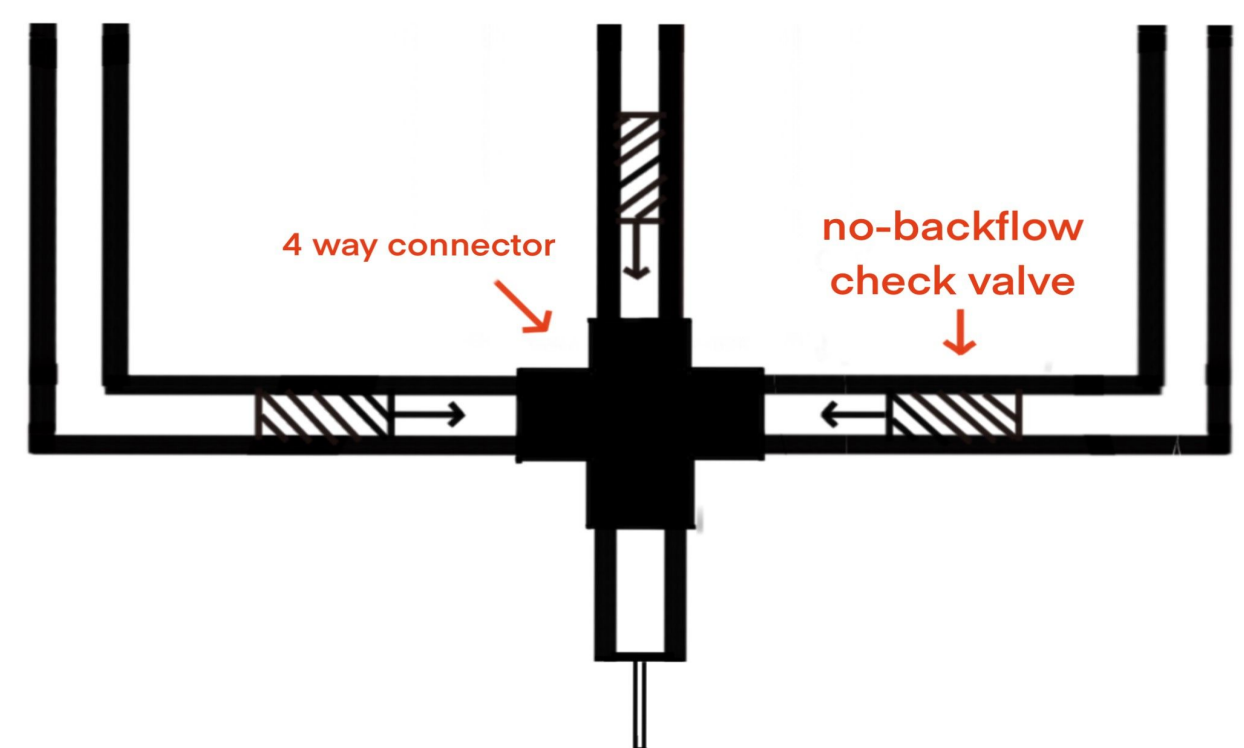


Figure 4: Diagram of check valve system at intersection

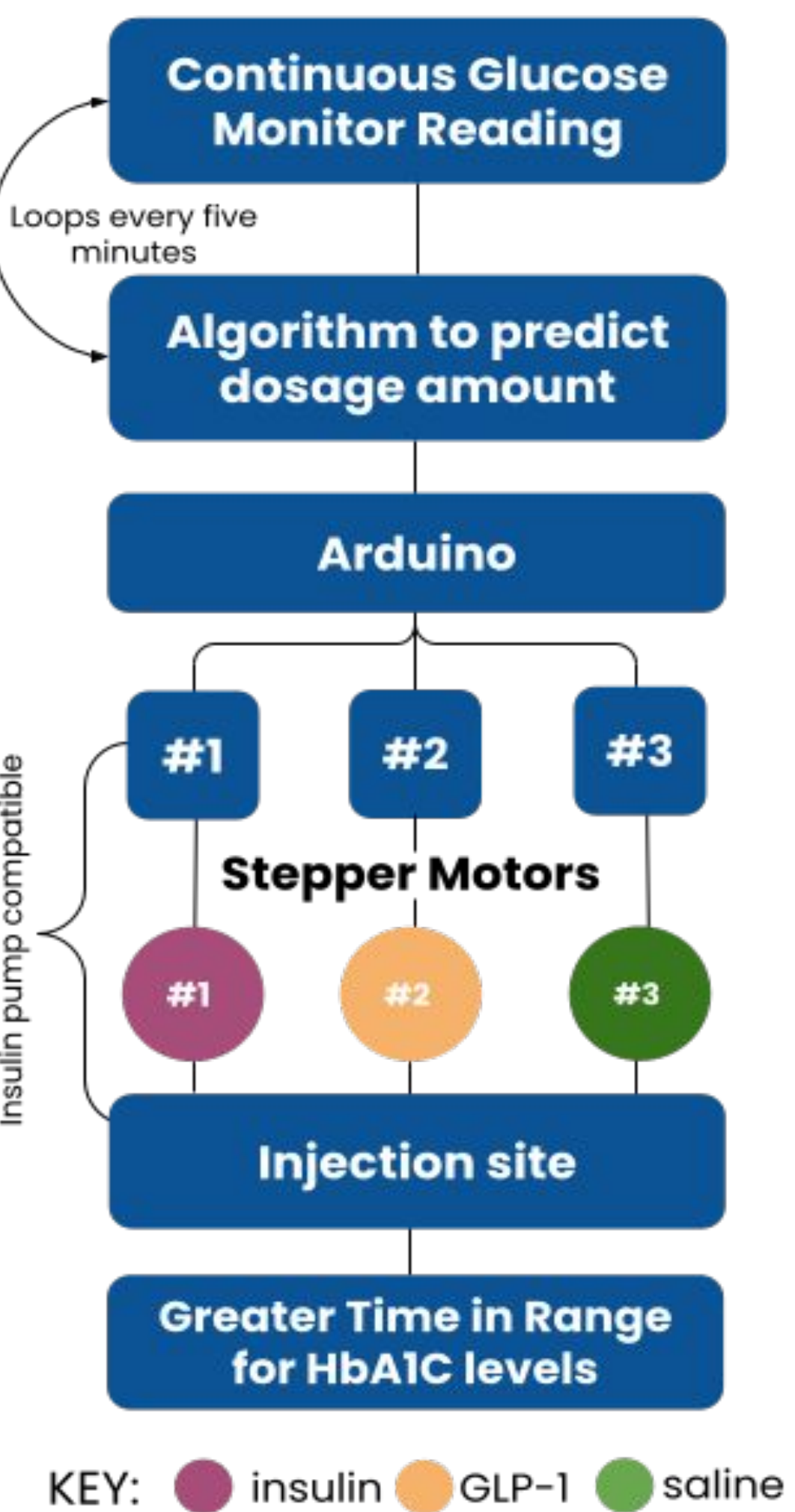


Figure 3: Process Flowchart

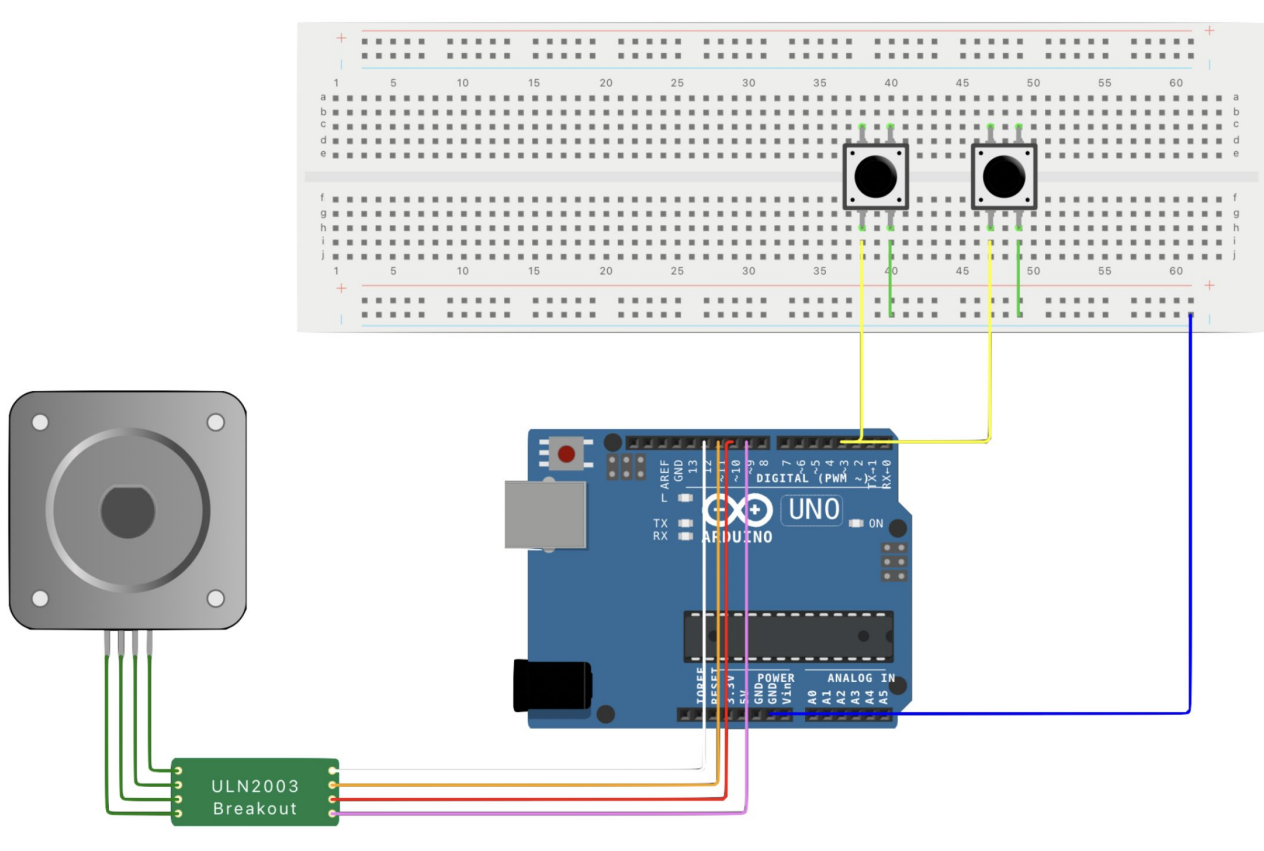


Figure 5: Circuit diagram of Arduino that controls one stepper motor

Our solution combines insulin and GLP-1 delivery through a rack and pinion system powered by an Arduino, delivering doses from pre-filled cartridges using CGM data. Saline solution rinses tubing after each use, and a slow flow rate ensures laminar flow. Our device also utilizes check valves to prevent backflow at the intersection.

Prototype Testing and Results

Motor Type:	Advantages:	Disadvantages:	Used?
DC Motor	High torque	Incapable of slower speeds	X
Microservo	High torque	Choppy movement	X
5V Stepper Motor	Smooth movement	Not powerful enough	X
12V Stepper Motor	High torque, smooth movement	---	✓

Figure 5: Table of different types of motors tested and benefits of each

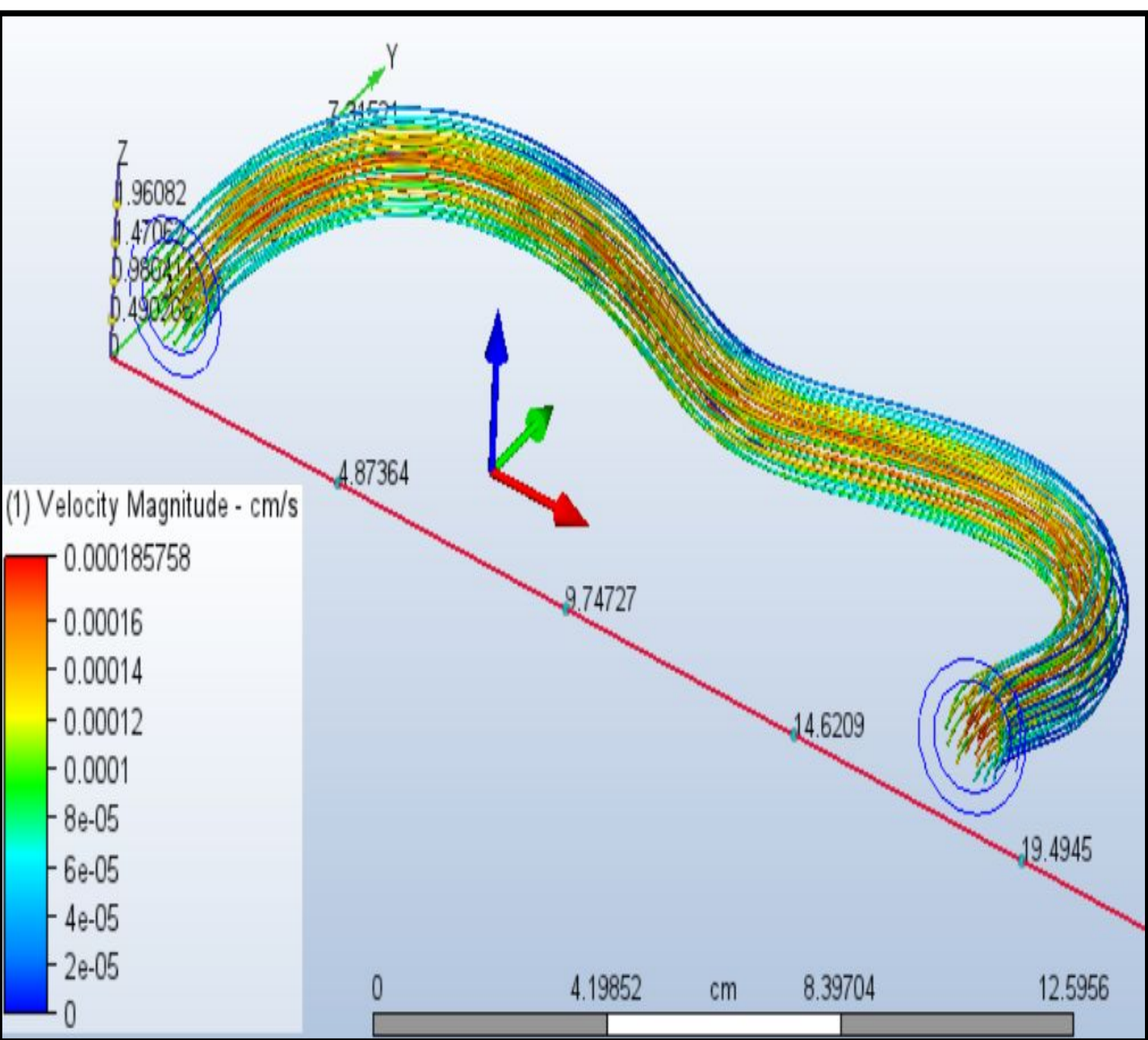


Figure 6: Simulation of GLP-1RA fluid flow through silicon rubber tubing demonstrating points of highest velocity

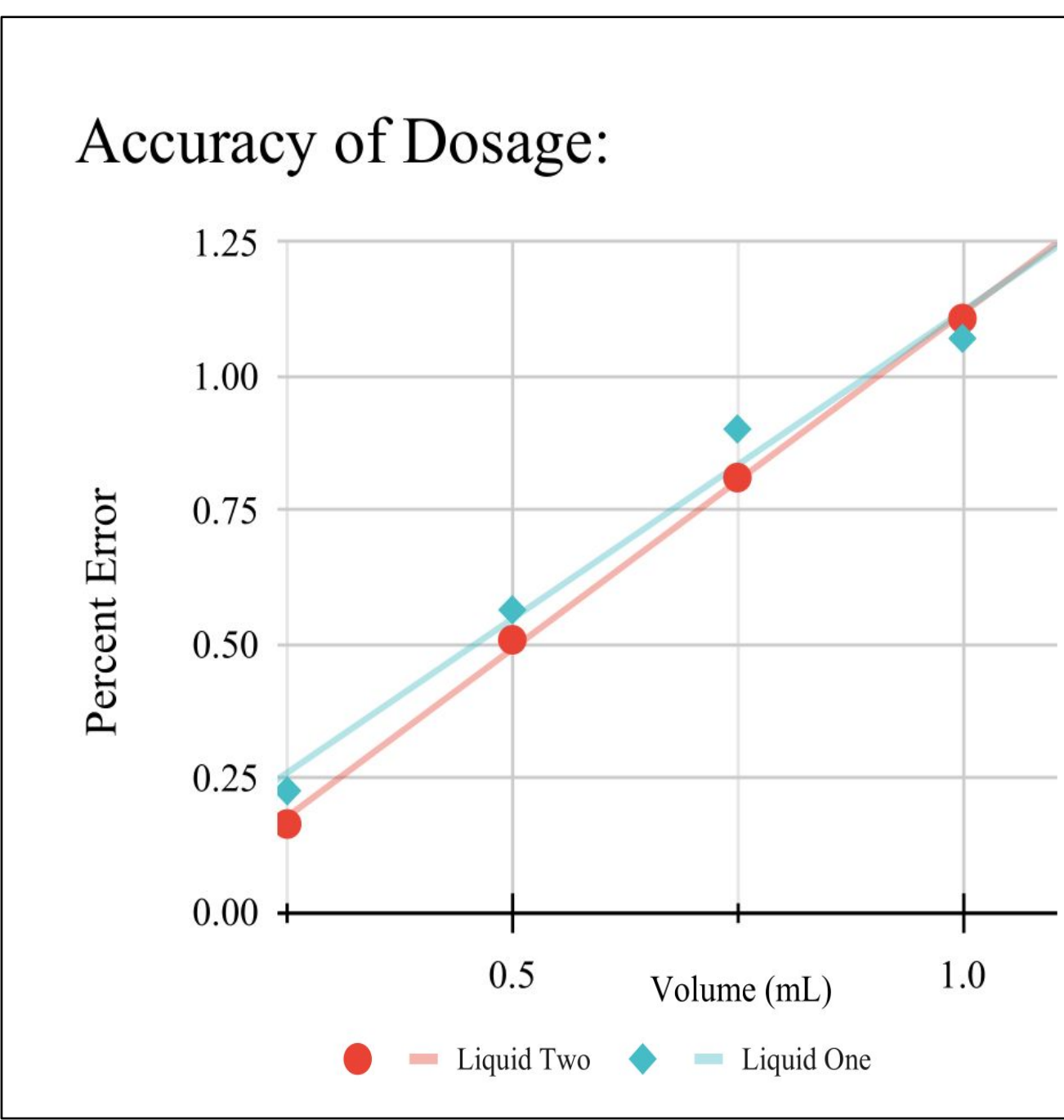


Figure 7: Accuracy graph for prototype, volume versus percent error

Next Steps

- Reduce the size of prototype to the size of an Omnipod or a similarly-sized product
- Integrate CGM data and machine learning to control dosage levels, allowing for less user input
- Continue testing to ensure accuracy of dosage
- Design algorithm that calculates dosages based on patient statistics

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